

Complete BCZ Farey Cycles: Physical Clocks and Boundary-Safe Parabolic Returns

HCS-C395 source-theorem and reproducibility package

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Abstract

We connect every primitive cycle of the classical Farey-triangle return to its least physical horocycle period and its complete branch cocycle. The discrete cycle length is a totient sum, whereas the physical period is the inverse square of its real scale. Summing successive Farey gaps proves the physical formula with no unrecorded rescaling. Transposing the lattice basis-update identity then determines a nontrivial unipotent return matrix at every starting point and all its powers. These formulas retain the full rational-layer classification and irrational exclusion. At included floor walls the map can be discontinuous: the exact branch cocycle remains meaningful, but a two-sided smooth return derivative is not asserted. At interior scales the derivative has two unit multipliers, as required by the continuous radial family. The result supplies a boundary-safe classical source ledger, with independently checked exact matrix and clock identities rather than a target determinant.

中文摘要

本文进一步连接每条本原法里循环的离散长度、物理横流周期和完整分支返回矩阵。离散长度是欧拉函数之和，物理周期则是尺度平方的倒数；法里间隙求和证明两者不能混用。格基更新恒等式确定每个起点的非平凡幂零剪切及全部重复。完整有理层分类与无理斜率排除仍然保留。包含的取整边界处映射可能不连续，因此只保留精确分支矩阵，不虚称双侧光滑导数。内部尺度处的返回导数具有两个单位乘子，与径向连续周期族一致。所有结论均属于经典源动力学，不构造目标行列式。

Keywords: Farey triangle; horocycle return; primitive cycles; parabolic cocycle; roof integrability; source boundary

关键词：法里三角形；横流返回；本原循环；抛物矩阵；顶函数可积性；源系统边界

1 One section, native arithmetic, and classical ownership

The presence of exact arithmetic in a dynamical system does not by itself give a discrete prime-bearing orbit ledger. The Farey-triangle return is a particularly explicit test: its integer pairs are intrinsic primitive lattice coordinates, not labels added after the dynamics is computed. This paper keeps the source object, its clocks and its boundary conventions fixed throughout the argument.

The section and periodic classification are classical results of Athreya and Cheung [1]. Their construction identifies the Boca–Cobeli–Zaharescu map as a horocycle first return and relates its rational layers to Farey fractions. We reconstruct the needed arguments with explicit endpoint and matrix conventions and an independent finite verification package. This is an owner-heavy source reconstruction, not a claim of new literature priority. Cheung and Quas later established weak mixing [2]; that theorem is acknowledged as background and is neither reproved nor used below. Strong mixing and a complete spectral type are not claims of this paper.

Set

$$\Omega = \{(a, b) : 0 < a, b \leq 1, a + b > 1\}, \quad T(a, b) = (b, kb - a), \quad k = \left\lfloor \frac{1+a}{b} \right\rfloor. \quad (1)$$

The roof is $R(a, b) = 1/(ab)$. The floor takes its integer value at equality; the lower edge is excluded and the upper edges are included. We use column vectors, with branch matrix $A_k = \begin{pmatrix} 0 & 1 \\ -1 & k \end{pmatrix}$. One application of T is one discrete return, not one unit of physical horocycle time.

2 Exact first return, inverse, and irrational exclusion

Let

$$p_{a,b} = \begin{pmatrix} a & b \\ 0 & a^{-1} \end{pmatrix}, \quad h_s = \begin{pmatrix} 1 & 0 \\ -s & 1 \end{pmatrix}, \quad \mathcal{L}_{a,b} = p_{a,b} \mathbb{Z}^2.$$

A lattice with a primitive positive horizontal generator of length $a \leq 1$ has a unique representation in this section: the second basis vector has height $1/a$, and its horizontal coordinate has a unique representative $b \in (1-a, 1]$ modulo $a\mathbb{Z}$. A lattice vector is $(x, n/a)$, where $x = am + bn$ and m, n are integers. At a positive horizontal-return time s , it obeys $0 < x \leq 1$ and $n/a = sx$, hence $n \geq 1$. If $s < 1/(ab)$, then $x > nb$, so $m \geq 1$ and $x \geq a + b > 1$, a contradiction. At $s = 1/(ab)$ the primitive vector $(b, 1/a)$ becomes $(b, 0)$. This proves the exact first return.

The new oriented basis is $(b, 0), (-a + kb, 1/b)$. The floor inequality gives $c = kb - a \leq 1 < c + b$, so $(b, c) \in \Omega$, including at equality. With $B_k = A_k^T$, the basis identity is

$$p_{T(a,b)} = h_{R(a,b)} p_{a,b} B_k, \quad B_k = \begin{pmatrix} 0 & -1 \\ 1 & k \end{pmatrix}. \quad (2)$$

We assert this first-return description for the stated section. We do not assert that this section exhausts every orbit of the whole lattice-space flow; the short closed horocycles omitted by the classical global section statement remain outside this claim.

If $T(a, b) = (b, c)$, then $(1 + c)/b = k + (1 - a)/b$ and $0 \leq (1 - a)/b < 1$. Thus

$$T^{-1}(a, b) = \left(\left\lfloor \frac{1+b}{a} \right\rfloor a - b, a \right).$$

The swap $J(a, b) = (b, a)$ therefore satisfies $JTJ = T^{-1}$ and $R \circ J = R$, with the same endpoint convention. Every open branch has determinant one. Its countably many walls have area zero, and the inverse shows that branch images partition Ω up to this null set. Consequently $d\mu = 2 da db$ is invariant probability.

Lemma 1. *Every periodic point of T has rational slope b/a .*

Proof. If $T^n(a, b) = (a, b)$ and $s_n = \sum_{j=0}^{n-1} R(T^j(a, b))$, iteration of (2) gives

$$B_{k_0} \cdots B_{k_{n-1}} = p_{a,b}^{-1} h_{-s_n} p_{a,b} = \begin{pmatrix} 1 - abs_n & -b^2 s_n \\ a^2 s_n & 1 + abs_n \end{pmatrix}. \quad (3)$$

The left side is integral. Hence $a^2 s_n$ is a positive integer and abs_n is an integer; their ratio is b/a . No continuity of the branch itinerary was used. $\square$

3 All rational layers and exact least periods

For $N \geq 1$, define

$$S_N = \{(q, r) : 1 \leq q, r \leq N, \gcd(q, r) = 1, q + r > N\}, \quad L_N = \sum_{j=1}^N \varphi(j), \quad \varphi(1) = 1.$$

Let $\mathcal{F}_N$ be the sorted reduced fractions in $[0, 1]$ with denominator at most N , including $0/1$ and $1/1$. There are $1 + L_N$ fractions, hence L_N gaps. Extend the list by integer translation when passing from the last gap to the first; the two endpoints are not two distinct marked points of the cyclic denominator list.

Lemma 2. *Consecutive denominator pairs in this cyclic list are exactly S_N , each once, and their successor is $(q, r) \mapsto (r, kr - q)$ with $k = \lfloor (N + q)/r \rfloor$.*

Proof. For determinant-one neighbors $p/q < s/r$, an intermediate fraction c/d satisfies

$$d = q(sd - rc) + r(qc - pd) \geq q + r.$$

Both parentheses are positive integers. This proves the absence of an intermediate denominator at most N whenever $q + r > N$. All neighbors have determinant one: induct from order one by inserting mediants exactly when the denominator sum is $N + 1$. Every new reduced c/d , $d = N + 1$, has these parents. Choose $q \in \{1, \dots, d - 1\}$ with $cq \equiv 1 \pmod{d}$, set $p = (cq - 1)/d$, $r = d - q$, and $s = c - p$. The parents lie in $[0, 1]$, have determinant one and denominator sum d , and the previous inequality proves they were neighbors at order N .

For a given $(q, r) \in S_N$, choose the unique $0 \leq p < q$ such that $pr \equiv -1 \pmod{q}$, setting $p = 0$ when $q = 1$. Then $s = (1 + pr)/q$ gives its unique neighboring fractions in $[0, 1]$. Finally $t = kr - q$ obeys $0 < t \leq N$ and $t + r > N$. The fraction $(ks - p)/t$ has determinant one with s/r ; it is the next neighbor in the translated list. This proves the successor formula. $\square$

Theorem 1. *Every rational-slope point is uniquely $(\delta q, \delta r)$ with positive coprime q, r . For $N = \lfloor 1/\delta \rfloor$ its membership in Ω is equivalent to*

$$\frac{1}{N+1} < \delta \leq \frac{1}{N}, \quad (q, r) \in S_N.$$

At each such scale, δS_N is exactly one primitive cycle of least period L_N . These are all periodic cycles of the map.

Proof. Write $1/\delta = N + \theta$, $0 \leq \theta < 1$. Integer division gives

$$\left\lfloor \frac{1 + \delta q}{\delta r} \right\rfloor = \left\lfloor \frac{N + q + \theta}{r} \right\rfloor = \left\lfloor \frac{N + q}{r} \right\rfloor,$$

since an integer remainder is at most $r - 1$. Lemma 2 then visits every denominator pair exactly once. No earlier return is possible. The scale and the reduced pair are unique; Lemma 1 excludes any other periodic points. $\square$

For clarity, $L_1 = 1$, $L_2 = 2$, $L_3 = 4$, and the third cycle is $(1, 3), (3, 2), (2, 3), (3, 1)$ before scaling. The lower scale endpoint belongs to the next order, not to this cycle. The map at the upper endpoint retains the exact floor values.

4 Physical return time and complete parabolic cocycle

Theorem 2. *The least positive horocycle period of the cycle at scale δ is δ^{-2} , and its complete branch cocycle starting at $(\delta q, \delta r)$ is*

$$M_{q,r} = \begin{pmatrix} 1 - qr & q^2 \\ -r^2 & 1 + qr \end{pmatrix}. \quad (4)$$

Both eigenvalues are one. If $D = M_{q,r} - I$, then $D \neq 0$, $D^2 = 0$, and $M_{q,r}^\ell = I + \ell D$ for all integers ℓ . Physical repetitions have $\ell \geq 1$ and total roof $\ell \delta^{-2}$.

Proof. A Farey gap is $s/r - p/q = 1/(qr)$. Summing all gaps of one unit interval gives

$$\sum_{(q,r) \in S_N} \frac{1}{qr} = 1, \quad \sum_{j=0}^{L_N-1} R(T^j(\delta q, \delta r)) = \delta^{-2}.$$

This is the least positive flow period, not merely a return period: integrality of $p_{a,b}^{-1} h_s p_{a,b}$ requires $s\delta^2 q^2, s\delta^2 qr, s\delta^2 r^2$ to be integers. Their integer coefficients have greatest common divisor one. Bezout's identity therefore forces $s\delta^2 \in \mathbb{Z}$. The positive value $s = \delta^{-2}$ works.

Transpose (3), use $s_{L_N} = \delta^{-2}$, and keep the last branch on the left. This gives $A_{k_{L_N-1}} \cdots A_{k_0} = M_{q,r}$. Finally $D = (q, r)^\top (-r, q)$ is nonzero, has rank one, and squares to zero. The binomial identity gives all powers, including algebraic negative powers; negative integers are not physical repeat periods. $\square$

Why the floor-wall distinction is essential. At strict interior scale $1/(N+1) < \delta < 1/N$, every floor argument has a nonzero fractional part, and every coordinate lies below one. The finite itinerary is constant in an open neighborhood, so (4) is the actual derivative of T^{L_N} . At $\delta = 1/N$, the orbit contains $(1, 1/N)$ and a floor argument $2N$. For example, $T(1, 1/2) = (1/2, 1)$, but

$$\lim_{\varepsilon \downarrow 0} T(1 - \varepsilon, 1/2 + \varepsilon) = (1/2, 1/2).$$

The map is discontinuous there. Its exactly specified branch product still satisfies (4); calling it a two-sided smooth derivative would be false. Even in the smooth interior case, $\det(I - M_{q,r}) = 0$ because the fixed points vary in a radial family. An isolated nondegenerate periodic-orbit trace formula cannot be imposed on that family.

A reversal is not an extra cycle quotient. The swap carries a cycle to its inverse-time presentation and preserves the roof. It acts within the fixed-scale cycle because the complete denominator set is invariant under swapping q, r . The primitive cycle remains a cycle with L_N marked points; it is not divided by two when this reversal is applied. The shear formula retains the specified starting pair, and cyclic changes of starting point conjugate the return matrix by the intervening branch matrices.

5 Independent evidence and scope

The canonical producer follows integer floor iteration from $(1, N)$. The independent checker imports no producer code: it constructs and sorts every reduced Farey fraction, compares every consecutive denominator pair, and verifies the floor inequalities, inverse recurrence and scalar lattice-return identity at an interior scale and at the included endpoint. All $N = 1, \dots, 64$ cycles contain 27,833 marked points in total. The two scales give 55,666 exact step controls. There are 27,833 starting-position return matrices, 320 positive repetition controls, 16 explicit floor-wall discontinuity controls and 128 fixed-iterate descriptions. These are complete finite populations, not an all-order proof by extrapolation.

The symbolic lane separately checks 14 universal matrix identities and 256 exact layer identities. Its 65 ninety-digit numerical controls are quadrature and asymptotic consistency checks, not certified enclosures. Two unrelated-directory producer replays are byte identical. Repaired-hash semantic attacks, strict literal-type and unknown-key checks, actual release-write rejection of malformed evaluation files, and refusal of optimized Python modes protect the declared evidence contract. Universal quantifiers are supplied by the proofs, not by these finite checks or by the release hash.

The source has intrinsic coprime arithmetic and complete primitive dynamics, but no natural correspondence from rational primes to primitive cycles of length $\log p$. Neither the totient sum nor the continuous physical roof supplies target amplitudes or signs. No target local arithmetic, Euler factors, root number, automorphy, divisor, functional equation, zero match or Hilbert–Polya operator is claimed. Route B remains disabled.

NO_BAD_EULER_OR_ROOT_NUMBER.

Round one: physical clocks and boundary-safe parabolic returns.

References

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